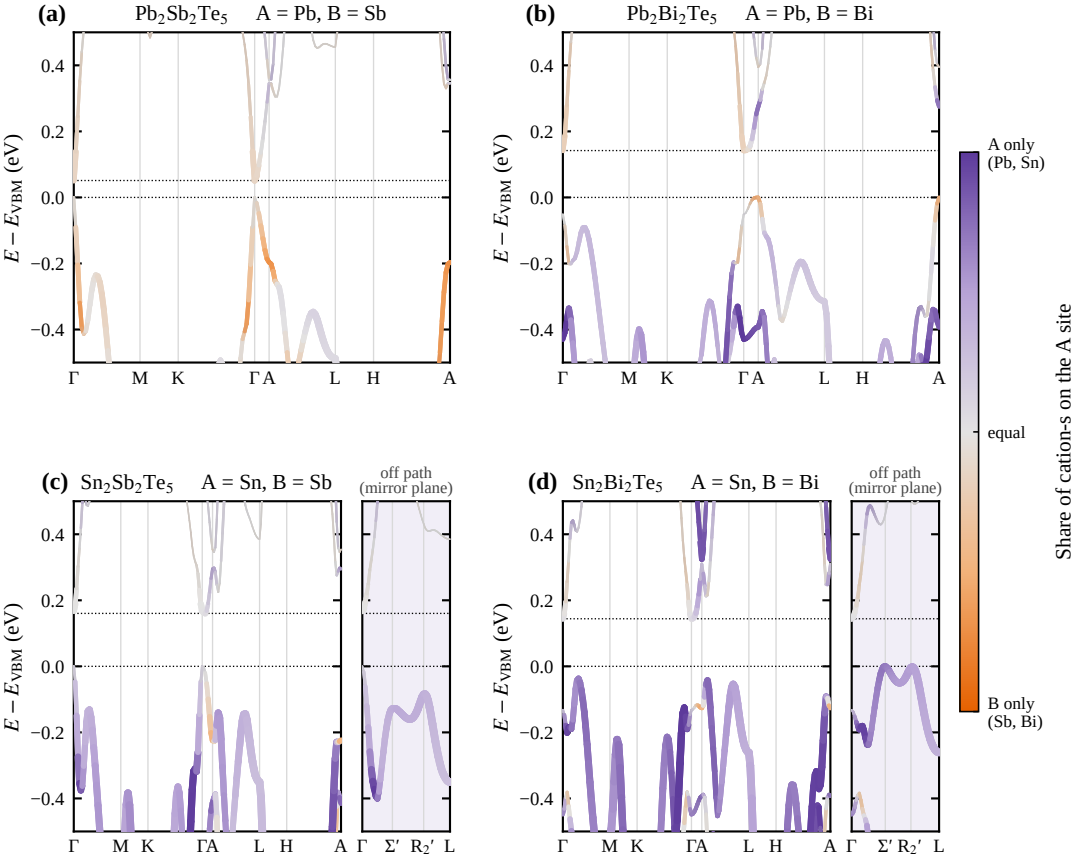

(a)
$Pb_2Sb_2Te_5$ A = Pb, B = Sb
(b)
$Pb_2Bi_2Te_5$ A = Pb, B = Bi
(c)
$Sn_2Sb_2Te_5$ A = Sn, B = Sb
off path (mirror plane)
(d)
$Sn_2Bi_2Te_5$ A = Sn, B = Bi
off path (mirror plane)
$E - E_{VBM}$ (eV)
0.4
0.2
0.0
−0.2
−0.4
Γ M K ΓA L H A
Γ Σ′ R$_2$′ L
A only (Pb, Sn)
equal
B only (Sb, Bi)
Share of cation-s on the A site